Upgrade Single Instance GI and Multiple DBs from 12c to 19c

Introduction: This readme helps to understand how to upgrade of **Single Instance Grid Infrastructure** along with **Multiple Oracle Databases on ASM/JFS** from **Oracle 12c to 19c using Ansible automation modules**. This module will do the installation of 19c Single Instance GI and upgrades it, then installs 19c RDBMS and upgrades the database using the Oracle's "autoupgrade.jar" tool. About Oracle Database AutoUpgrade.

Below figures provide a pictorial representation of the upgrade process using this Ansible collection.

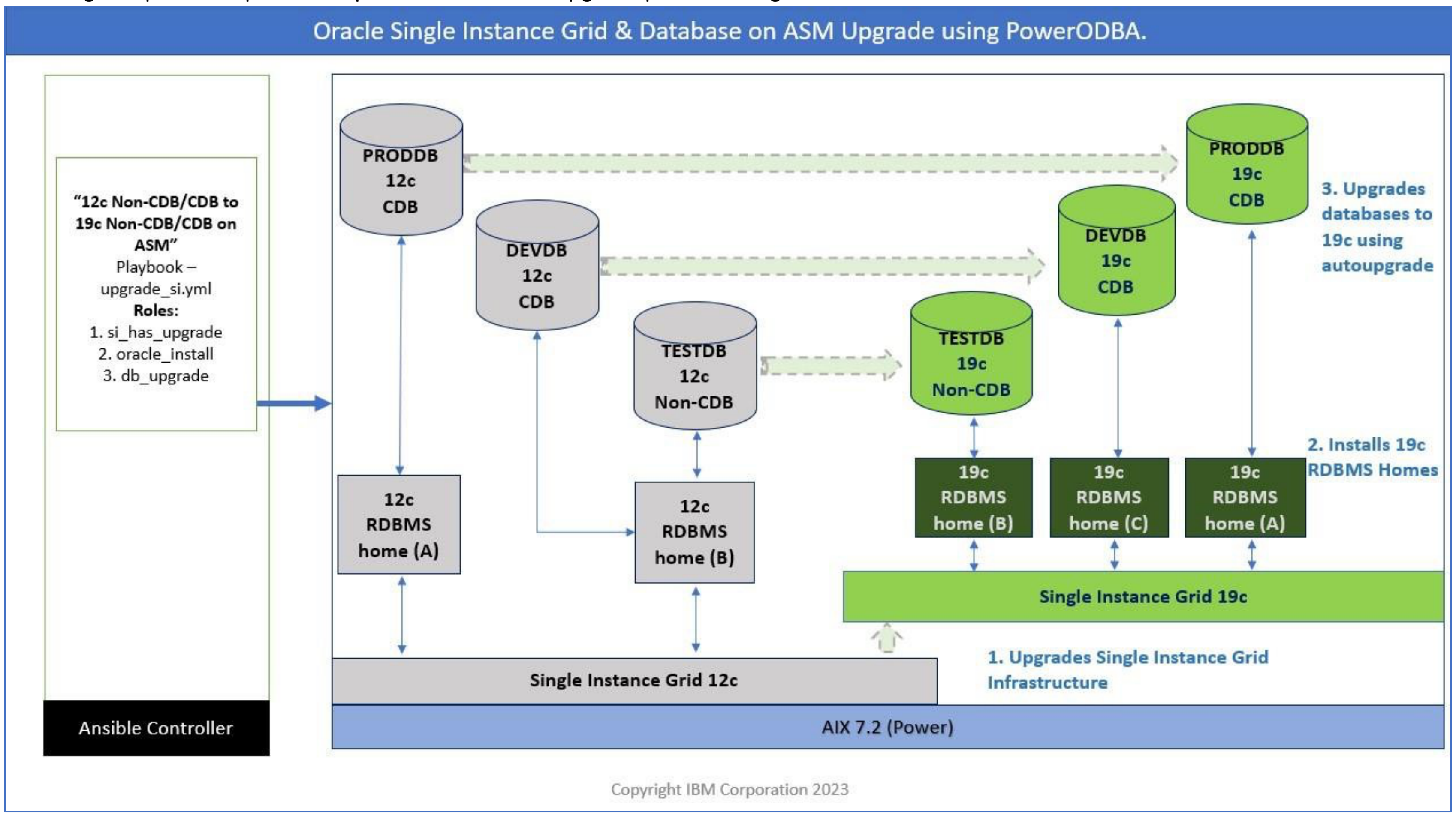

Fig: 1 Upgrading 12c Non-CDB/CDB to 19c Non-CDB/CDB

Note: All the pluggable databases inside the container database will be upgraded from 12c to 19c as part of upgrade process.

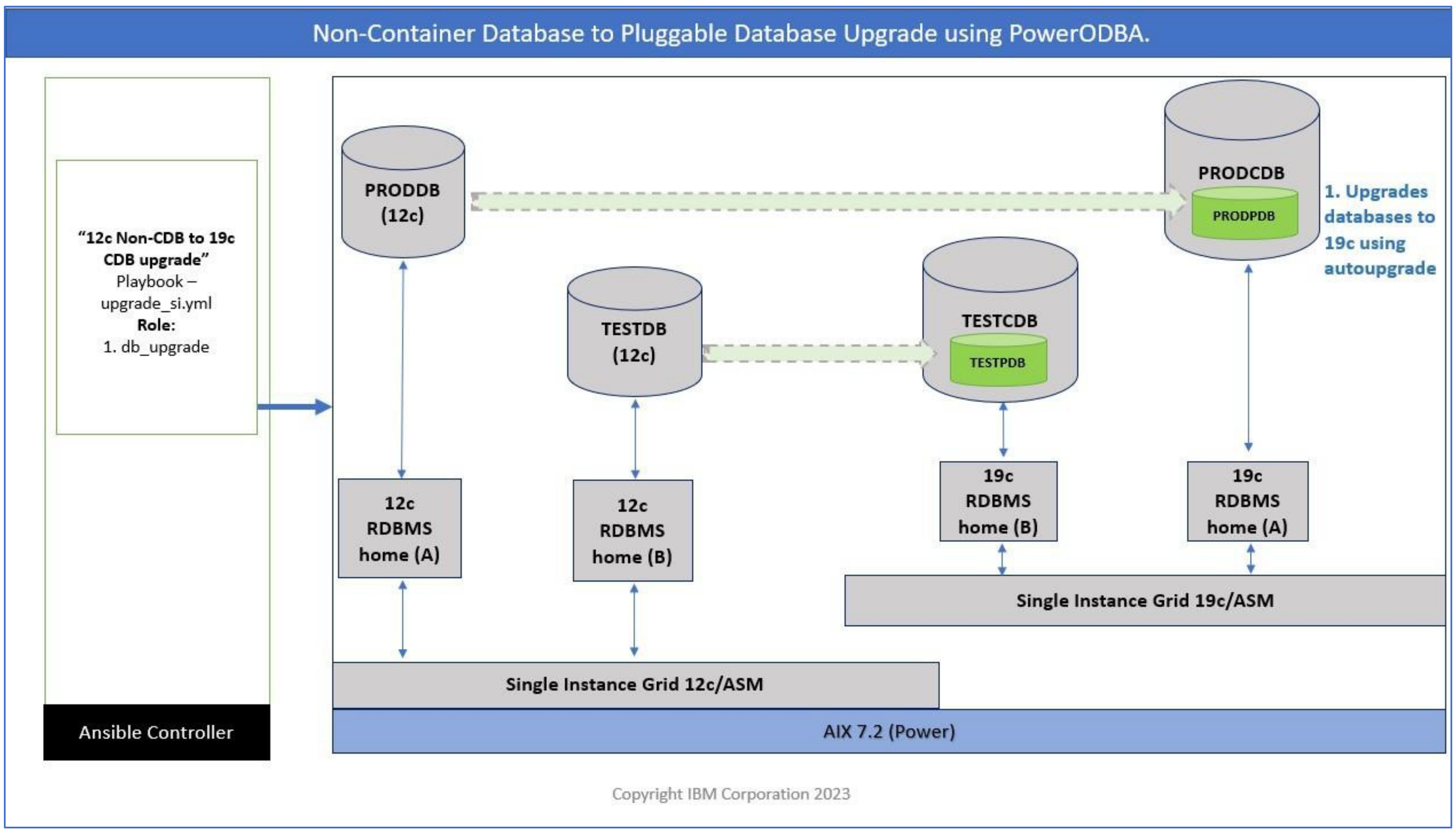

Fig: 2 Upgrading 12c Non-CDB to 19c CDB

Note: When upgrading from 12c Non CDB to 19c CDB, the 19c Container database should be created manually and the variables file should be updated before running the playbooks. Please read "Other scenarios" of "Upgrade Scenarios" provided in the later section of this readme.

File system architecture:

1. **Path to the collection**: $*ansible-collection-install-dir*/ibm/power_aix_oracle_dba.
2. **upgrade_si.yml**: This is the *playbook* file which is responsible for installation and upgrade of 19c Single instance Grid & databases by calling the respective roles. The file is under - ***"ansible-collection-install-dir"*/ibm/power_aix_oracle_dba/playbooks**. Only the managed host's hostname must be updated in this file.
3. **inventory.yml**: This file is provided in the collection which contain all the managed hosts details. It is NOT mandatory to use only this file, if you already have an inventory file defined in another location, that can be used.
4. **upgrade_si_vars.yml**: This file contains all the variables required to perform the upgrade. It is under is under - ***"ansible-collection-installdir"*/ibm/power_aix_oracle_dba/playbooks/vars/upgrade**. Specification of each variable is provided in this file itself.
5. **vault.yml**: The sys user password of ASM must be mentioned in this file, this file is in ***"ansible-collection-installdir"*/ibm/power_aix_oracle_dba/playbooks/vars.** It must be encrypted using "ansible-vault" after the password is stored in the file. Ansible Vault is a security utility provided by Ansible to encrypt files which contain sensitive information such as passwords. Refer: [A brief introduction to Ansible Vault | Enable Sysadmin (redhat.com)]

   $ ansible-vault encrypt vault.yml

6. **Roles**: There are three ansible roles which will be used to perform the upgrade. Description of each role is given below.

Ansible Roles to Upgrade SI GI & DBs		
si_has_upgrade	**oracle_install**	**db_upgrade**
Upgrades Single Instance 12c Grid to 19c with (or) without RU.	Installs 19c Oracle database home with (or) without RU for DB upgrades.	Upgrades the database using autoupgrade.jar utility.
1. Checks if 19c Single Instance Grid Infrastructure is already installed. 2. Checks for source (12c) grid setup. 3. Checks the required value of maxuproc value. 4. Checks for mandatory patch on source (12c) grid home for 19c upgrade. 5. Extracts the 19c Grid software. 6. Backups up OPatch and extracts latest OPatch when apply_ru is set to True. 7. Extracts Release update patch when apply_ru is set to True. 8. Checks freespace in Grid home path and fails if freespace is less than 80GB, it is mandatory check set by Oracle to have 80GB of freespace to do the patching. 9. Runs cluvfy.sh and pauses the task in case any missing/failed items are found. 10. Executes rootpre.sh 11. Creates & copies grid_upgrade.rsp response file from template to the target lpar. 12. Executes gridSetup.sh (with upgrade option) in silent mode along with -applyRU when apply_ru is set to True or executes without applyRU option when apply_ru is set to False. It will show the status and lists the log files upon completion of the new 19c Grid Installation. 13. **Upgrade:** Stops the databases on 12c DB homes which are provided for *source_db_name* in the *databases* section of the variables file (upgrade_si_vars.yml). 14. Executes rootupgrade.sh. 15. Executes gridSetup.sh -silent -executeConfigTools and lists the log files upon completion. 16. Checks and displays the status of grid services. 17. Starts the databases from 12c DB home which were stopped before the upgrade.	1. Checks for any failed 19c RDBMS installations. 2. Extracts the 19c RDBMS software. 3. Backups up OPatch and extracts latest OPatch when apply_ru is set to true. 4. Extracts Release update patch when apply_ru is set to True. 5. Executes rootpre.sh 6. Creates & copies oracle_install.rsp response file from template to the target lpar. 7. Installs 19c RDBMS software. 8. Executes root.sh 9. Continues to create multiple Oracle homes based on the number of "target_db_home" variables provided in the variables file.	1. Creates a stage directory to place the autoupgrade configuration file. 2. Checks DB name in oratab. 3. Checks DB is up and running or not. 4. Generates autoupgrade configuration file based on the inputs provided in the variables file. 5. Executes autoupgrade in analyze mode. [User must review the "analyze" results and fix them before running the "deploy" mode. 6. User should execute the playbook with deploy tag for autoupgrade to run in deploy mode.

Important Steps to know before starting the upgrade automation:

1. Apply required patches on the 12c environment to avoid unintended errors prior to the upgrade process. Refer MOS Doc ID 2539751.1.
2. Stage the 19c Grid home software and 19c RDBMS home software, Release Update (RU) patch & latest Opatch utility zip files in a specific directory on the target lpar and mention that path for the variable "sw_stage" in the the variables file [power_aix_oracle_dba/playbooks/vars/upgrade/upgrade_si_vars.yml].
3. Always use the latest autoupgrade.jar file [AutoUpgrade Tool (Doc ID 2485457.1)] and stage that in a directory on the target lpar and mention that path for the variable "autoupgrade_file_loc" in the variables file [power_aix_oracle_dba/playbooks/vars/upgrade/upgrade_si_vars.yml].
4. The freespace in the installer path of 19c homes should be more than 80GB. This is applicable when you want to apply Release Update patches (RU) along with Grid & Oracle Homes installation.
5. If you want the autoupgrade to create a restore point, enable the Flashback mode, and maintain sufficient space in Flash Recovery Area (FRA) to avoid failure during the upgrade process.
   SQL> alter system set db_recovery_file_dest_size=35G; -- Increase the size as per your environment.
   SQL> alter system set db_recovery_file_dest='+FRA'; -- For example, FRA diskgroup was used. JFS path can also be used.
   SQL> shutdown immediate;
   SQL> startup mount;
   SQL> alter database archivelog;
   SQL> alter database flashback on;
   SQL> alter database open;
6. To proceed with the upgrade without creating a restore point, update the variable "restoration" in the "databases" section to 'no' in the variables file power_aix_oracle_dba/playbooks/vars/upgrade/upgrade_si_vars.yml
7. The list of 12c databases running on ASM must be mentioned in the variable "source_db_name" of the "databases" section in power_aix_oracle_dba/playbooks/vars/upgrade/upgrade_si_vars.yml file. The listed databases will be shut down during Single Instance Grid Infrastructure (rootupgrade.sh) upgrade.
8. Please use your standard backup strategy before starting the upgrade process. This playbook won't backup the databases.
9. Upgrade of Grid Infrastructure for RAC & RAC databases are NOT SUPPORTED.
10. Always run the playbooks in a vnc viewer to avoid ssh timeouts.
11. Following tags are provided:
    a. si_has_upgrade: This will invoke the role "si_has_upgrade". Upgrades SI Grid Infrastructure to 19c from 12c.
    b. oracle_install: This will invoke the role "oracle_install". Installs 19c Oracle Homes for database upgrades from 12c to 19c.
    c. predbupgrade: This will invoke prerequisite part of "db_upgrade" role which will do prechecks on the existing databases running on the lpar.
    d. analyze: This will invoke "autoupgrade" analyze mode which is a section of "db_upgrade" role.
    e. deploy: This will invoke a section of "db_upgrade" role which will invoke "autoupgrade" deploy mode, which is the core database upgrade mode.
12. These playbooks will create three directories (podba*) in /tmp. They should NOT be removed until the upgrade process completes otherwise it will compromise idempotency.
13. Try this on a Non-production environment first before using it on a Production environment.

Upgrade Scenarios:

1. **Full Stack Upgrade**: Upgrade the Single Instance GI & all the databases running on ASM. Playbook must be executed twice as shown below. The following command will perform 19c Single instance GI installation and upgrade, installation of 19c RDBMS along with Autoupgrade Analyze mode.
   - ansible-playbook upgrade_si.yml -i inventory.yml --ask-vault-pass –tags si_upgrade,oracle_install,predbupgrade,analyze

   Users must review the logs and fix any errors/recommendations reported by the autoupgrade tool and rerun the playbook with "deploy" tag.
   - ansible-playbook upgrade_si.yml -i inventory.yml --ask-vault-pass --tags deploy
2. **Other scenarios:**
   a) To upgrade only Single Instance Grid, following command must be executed.
      - ansible-playbook upgrade_si.yml -i inventory.yml --ask-vault-pass --tags si_upgrade

   b) If the databases are running on JFS then Single Instance Grid upgrade is not required. In such case, skip the role "si_has_upgrade"
      - ansible-playbook upgrade_si.yml -i inventory.yml --ask-vault-pass --tags oracle_install,predbupgrade,analyze

   c) If 19c Oracle Homes are already installed, skip the role "oracle_install" and directly run the "db_upgrade" role.
      - ansible-playbook upgrade_si.yml -i inventory.yml --ask-vault-pass --tags predbupgrade,analyze
        Review the results of analyze mode of autoupgrade and execute "deploy" mode.
      - ansible-playbook upgrade_si.yml -i inventory.yml --ask-vault-pass --tags deploy.

   d) To upgrade a Non-Container database and plug it into a 19c Container database (upgrade & plug-in)
      - ansible-playbook upgrade_si.yml -i inventory.yml --ask-vault-pass --tags oracle_install
        Manually create a 19c container database and update the database variables required to do the upgrade.
      - ansible-playbook upgrade_si.yml -i inventory.yml --ask-vault-pass --tags predbupgrade,analyze
        Review the results of analyze mode of autoupgrade and execute "deploy" mode
      - ansible-playbook upgrade_si.yml -i inventory.yml --ask-vault-pass --tags deploy.

Playbook execution: In the following example, we are going to upgrade the Oracle 12.1.0.2 Single Instance GI to Oracle 19.26 along with one database. The s/w binaries and patches are placed in the local lpar. The Ansible playbook will extract them before doing the upgrade.

Playbook contents:

```
$ cat upgrade_si.yml
- hosts:         pilotdb
  remote_user: root
  gather_facts: False
  vars_files:
   - vars/upgrade/upgrade_si_vars.yml
   - vars/vault.yml
  roles:
   - role: podba_si_has_upgrade
     tags: si_has_upgrade
  tasks:
   - name: Import oracle_install role
     ansible.builtin.import_role:
       name: ibm.power_aix_oracle.oracle_install_db
       tasks_from: oracle_install.yml
     tags: oracle_install

   - name: Run podba_db_upgrade role
     ansible.builtin.import_role:
       name: podba_db_upgrade
```

Update the variables file upgrade_si_vars.yml:

```
# Section A - Update the Common Variables.
work_dir: /tmp/ansible
ora_binary_location: local
ora_nfs_host: 192.140.176.129
ora_nfs_device: /repos
ora_nfs_filesystem: /repos
grid_sw: /repos/images/oracle/19c/V982588-01_193000_grid.zip
database_sw: /repos/images/oracle/19c/V982583-01_193000_db.zip
apply_ru: True
ru_stage: /stage/19RU
opatch_file: /repos/images/oracle/opatch/12.2.0.1.44/p6880880_121010_AIX64-5L.zip
ru_file: /repos/images/oracle/19c/RU19.26/p37257886_190000_AIX64-5L_GI_RU19.26.zip
autoupgrade_util_remote: /home/ansible/binaries/autoupgrade.jar
autoupgrade_util: /home/oracle/autoupgrade.jar
ora_inventory: /u01/app/oraInventory
mgmt_opt:
oms_host:
oms_port:
oms_em_user:
ora_oinstall_group: oinstall
ora_group: dba

ora_base: /u02/app/base
ignoreprecheck: False

grid_asm_flag: True

# Section B - Update the Single Instance Grid Variables
grid_home_prev: /u01/grid12c/home
grid_ora_home: /u02/grid19c/home
ora_grid_user: grid
gi_osasm_group: oinstall
gi_osoper_group:
gi_cluster_name:

# Section C - Update the Variables for Oracle 19c RDBMS installation
ora_user: oracle
db_osoper_dba:
db_osbkup_dba:
db_osdg_dba:
db_oskm_dba:
db_osrac_dba:

# Section D - Variables for Databases.  [Please Don't Change the dictionary list format. This format is referenced in the code.]

autoupgrade_stage: /home/oracle
global_log_dir: /u02/autoupgrade
databases:
    - source_db_name: podba
      source_db_home: /u01/db12c/home
      ora_home: /u02/db19c/home
      log_dir: "{{ global_log_dir }}/dbupgrdlogs"
      start_time: NOW
      restoration: 'no'
      upgrade_node: localhost
      run_utlrp: 'yes'
      timezone_upg: 'yes'
      target_cdb_name:
      target_pdb_name:
```

Update the vaults.yml file with ASM sys password and encrypt it with ansible vault

```
bash-5.1$ cat vars/vault.yml

asm_password: Oracle4u # ASM sys user password for Single Instance GI Upgrade.

bash-5.1$ansible-vault encrypt vars/vault.yml
```

SIHA Version before the upgrade:

```
bash-5.1$ crsctl query has releaseversion
Oracle High Availability Services release version on the local node is [12.1.0.2.0]
bash-5.1$ hostname
pilotdb
```

Reference Output of SI Upgrade, Oracle Install, PreDB Upgrade & Analyze tasks:

```
[ansible@localhost playbooks]$ ansible-playbook upgrade_si.yml -i inventory.yml --ask-vault-pass --tags si_upgrade,oracle_install,predbupgrade,analyze
PLAY [pilotdb] *******************************************************************

TASK [Gathering Facts] ************************************************************
ok: [pilotdb]

TASK [Run initialization tasks] ***************************************************

TASK [ibm.power_aix_oracle.preconfig : Creating Ansible Work Directories] ******
changed: [pilotdb] => (item=/tmp/ansible)
changed: [pilotdb] => (item=/tmp/ansible/scripts)
changed: [pilotdb] => (item=/tmp/ansible/logs)
changed: [pilotdb] => (item=/tmp/ansible/done)
changed: [pilotdb] => (item=/tmp/ansible/files)
changed: [pilotdb] => (item=/tmp/ansible/backup)

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Checking if Upgrade Task was already done] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Fail if Upgrade Task was already done] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Checking if 19c GI is installed] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Checking if prechecks script is already run] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Checking current HAS Version] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Setting fact - HAS Version] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Templating out prechecks.sh script] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Executing prechecks.sh script] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Prechecks Script Output] ***
ok: [pilotdb] => {
    "precheck_out.stdout_lines": [
        "Prechecks completed successfully."
    ]
}

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Templating out Grid Response File] ***
changed: [pilotdb]

TASK [Include prep_gi_install.yml] ************************************************

TASK [ibm.power_aix_oracle.oracle_install_gi : Creating Oracle Base directory] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Creating Oracle Grid Home directory] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Checking if Grid S/W is extracted] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Checking if Opatch is extracted] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Checking if RU is extracted] ****
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Creating RU Stage Directory] ****
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Set Fact Target hostname for remote copy] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Creating NFS filesystem from nfshost.] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Extracting RU files from local/nfs] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Creating the done file] *********
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Copy RU files from remote] ******
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Extracting RU files] ************
```

```
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Creating the done file] *********
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Removing /tmp/ansible/files/p37257886_190000_AIX64-5L_GI_RU19.26.zip] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Extracting oracle grid source files from local/nfs] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Creating the done file] *********
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Copy GI sw files from remote] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Extracting GI sw files] *********
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Creating the done file] *********
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Removing /tmp/ansible/files/V982588-01_193000_grid.zip] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Backup Opatch] ******************
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Copy OPatch from remote] ********
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Extracting OPatch to /u02/grid19c/home] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Creating the done file] *********
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Removing /tmp/ansible/files/p6880880_121010_AIX64-5L.zip] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Backup Opatch] ******************
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Extracting OPatch from local/nfs to /u02/grid19c/home] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Creating the done file] *********
changed: [pilotdb]

TASK [Include grid_install.yml] ************************************************

TASK [ibm.power_aix_oracle.oracle_install_gi : Checking if 19c Grid is installed] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Checking freespace in Grid Installation Path /u02/grid19c/home] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Fail if the space is insufficient in /u02/grid19c/home] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Template out myruncluvfy.sh] ****
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Run myruncluvfy.sh] *************
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Read Cluvfy output file] ********
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Decode the file content] ********
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Fail if cluvfy reports any errors] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Get Patch ID] *******************
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Set Patch ID] *******************
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Executing rootpre.sh] ***********
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Templating out grid_install.sh Script] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Installation of Standalone Grid Infrastructure] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : debug] **************************
ok: [pilotdb] => {
    "grid_install_out.stdout": "INFO: gridSetup.sh completed successfully."
}

TASK [ibm.power_aix_oracle.oracle_install_gi : Templating out apply_ru_gi.sh Script] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_gi : Applying the patch] *************
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Checking if rootupgrade.sh script was already executed] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Stopping database services before running rootupgrade.sh] ***
changed: [pilotdb] => (item=None)
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Templating out grid_upgrade.sh (rootupgrade.sh) script] ***
changed: [pilotdb]
```

```
TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Executing grid_upgrade.sh script] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : ansible.builtin.debug] ***
ok: [pilotdb] => {
    "grid_upgrade_out.stdout": "Rootupgrade.sh completed successfully"
}

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Post Grid Upgrade Steps | Copying config_tools.sh] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Post GI Upgrade Steps| Executing the Config Tools script] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : ansible.builtin.debug] ***
ok: [pilotdb] => {
    "config_tools_out.stdout": "gridSetup.sh -executeConfigTools completed successfully"
}

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Post GI Upgrade Steps | Checking if Config Tools is successful.] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Removing Stale Files] ****
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Post GI Upgrade Steps| Status of GI Services] ***
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : ansible.builtin.debug] ***
ok: [pilotdb] => {
    "check_services.stdout_lines": [
        "NAME=ora.cssd",
        "TYPE=ora.cssd.type",
        "TARGET=ONLINE",
        "STATE=ONLINE on pilotdb"
    ]
}

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Post GI Upgrade Steps | Starting the database services] ***
changed: [pilotdb] => (item=None)
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Checking the status of the DBs] ***
changed: [pilotdb] => (item=None)
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_si_has_upgrade : Status of the DBs] *******
ok: [pilotdb] => (item=podba) => {
    "msg": " Status: podba Database is running."
}

TASK [ibm.power_aix_oracle.preconfig : Creating Ansible Work Directories] ******
ok: [pilotdb] => (item=/tmp/ansible)
ok: [pilotdb] => (item=/tmp/ansible/scripts)
ok: [pilotdb] => (item=/tmp/ansible/logs)
ok: [pilotdb] => (item=/tmp/ansible/done)
ok: [pilotdb] => (item=/tmp/ansible/files)
ok: [pilotdb] => (item=/tmp/ansible/backup)

TASK [ibm.power_aix_oracle.oracle_install_db : Check if inventory.xml exists] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Read Oracle Inventory File if it exists] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Decode inventory.xml content if available] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Prepare list of Oracle Homes for installation] ***
ok: [pilotdb] => (item={'source_db_name': 'podba', 'source_db_home': '/u01/db12c/home', 'ora_home': '/u02/db19c/home', 'log_dir': '/u02/autoupgrade/dbupgrdlogs', 'start_time': 'NOW', 'restoration': 'no',
'upgrade_node': 'localhost', 'run_utlrp': 'yes', 'timezone_upg': 'yes', 'target_cdb_name': None, 'target_pdb_name': None})

TASK [ibm.power_aix_oracle.oracle_install_db : Preparing a list of Oracle Homes defined in the Variables file] ***
ok: [pilotdb] => (item={'source_db_name': 'podba', 'source_db_home': '/u01/db12c/home', 'ora_home': '/u02/db19c/home', 'log_dir': '/u02/autoupgrade/dbupgrdlogs', 'start_time': 'NOW', 'restoration': 'no',
'upgrade_node': 'localhost', 'run_utlrp': 'yes', 'timezone_upg': 'yes', 'target_cdb_name': None, 'target_pdb_name': None})

TASK [ibm.power_aix_oracle.oracle_install_db : Checking if DB S/W is extracted] ***
ok: [pilotdb] => (item={'oh': 'u02db19chome', 'oracle_home': '/u02/db19c/home'})

TASK [ibm.power_aix_oracle.oracle_install_db : Checking if root.sh is run] *****
ok: [pilotdb] => (item={'oh': 'u02db19chome', 'oracle_home': '/u02/db19c/home'})

TASK [ibm.power_aix_oracle.oracle_install_db : Checking if OPatch is extracted] ***
ok: [pilotdb] => (item={'oh': 'u02db19chome', 'oracle_home': '/u02/db19c/home'})

TASK [ibm.power_aix_oracle.oracle_install_db : Get Patch ID] *******************
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Set Patch ID] *******************
ok: [pilotdb]

TASK [Including prep_db_install.yml task] **************************************

TASK [ibm.power_aix_oracle.oracle_install_db : Set Fact Target hostname for remote copy] ***
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Creating NFS filesystem from nfshost.] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Creating DB Home Directory] *****
changed: [pilotdb] => (item={'oh': 'u02db19chome', 'oracle_home': '/u02/db19c/home'})

TASK [ibm.power_aix_oracle.oracle_install_db : Extracting DB S/W from local/nfs] ***
changed: [pilotdb] => (item={'oh': 'u02db19chome', 'oracle_home': '/u02/db19c/home'})

TASK [ibm.power_aix_oracle.oracle_install_db : Copy DB S/W from remote] ********
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Extracting DB S/W files (Remote)] ***
skipping: [pilotdb] => (item={'oh': 'u02db19chome', 'oracle_home': '/u02/db19c/home'})
```

```
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Removing /tmp/ansible/files/V982583-01_193000_db.zip] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Get Patch ID] ******************
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Set Patch ID] ******************
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Checking if RU is extracted] ****
ok: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Creating RU Stage Directory] ****
changed: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Extracting RU files from local/nfs] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Creating the done file] *********
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Copy RU files from remote] ******
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Extracting RU files] ************
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Creating the done file] *********
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Removing /tmp/ansible/files/p37257886_190000_AIX64-5L_GI_RU19.26.zip] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Backup Opatch] ******************
changed: [pilotdb] => (item=/u02/db19c/home)

TASK [ibm.power_aix_oracle.oracle_install_db : Copy OPatch from remote] ********
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Extracting OPatch into Oracle Homes (Remote)] ***
skipping: [pilotdb] => (item=/u02/db19c/home)
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Removing /tmp/ansible/files/p6880880_121010_AIX64-5L.zip] ***
skipping: [pilotdb]

TASK [ibm.power_aix_oracle.oracle_install_db : Extracting OPatch from local|nfs] ***
changed: [pilotdb] => (item=/u02/db19c/home)

TASK [ibm.power_aix_oracle.oracle_install_db : Copying Oracle RDBMS Install response file] ***
changed: [pilotdb] => (item={'oh': 'u02db19chome', 'oracle_home': '/u02/db19c/home'})

TASK [ibm.power_aix_oracle.oracle_install_db : Copying oracle_install.sh] ******
changed: [pilotdb] => (item={'oh': 'u02db19chome', 'oracle_home': '/u02/db19c/home'})

TASK [ibm.power_aix_oracle.oracle_install_db : Installing 19c RDBMS] ***********
changed: [pilotdb] => (item={'oh': 'u02db19chome', 'oracle_home': '/u02/db19c/home'})

TASK [ibm.power_aix_oracle.oracle_install_db : Executing root.sh] **************
changed: [pilotdb] => (item={'oh': 'u02db19chome', 'oracle_home': '/u02/db19c/home'})

TASK [ibm.power_aix_oracle.oracle_install_db : Copying patch_install.sh] *******
changed: [pilotdb] => (item={'oh': 'u02db19chome', 'oracle_home': '/u02/db19c/home'})

TASK [ibm.power_aix_oracle.oracle_install_db : Running patch_install.sh] *******
changed: [pilotdb] => (item={'oh': 'u02db19chome', 'oracle_home': '/u02/db19c/home'})

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Stage Autoupgrade Utility] ***
skipping: [pilotdb]

TASK [Run initialization tasks] ************************************************

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Autoupgrade Full DB | Checking source database name in /etc/oratab file] ***
changed: [pilotdb] => (item=None)
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Fail if source database name doesn't exist in oratab file.] ***
skipping: [pilotdb] => (item=None)
skipping: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Autoupgrade Full DB | Checking the status of the Source Database] ***
changed: [pilotdb] => (item=None)
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Autoupgrade Full DB | Fail if Source database is not running.] ***
skipping: [pilotdb] => (item=None)
skipping: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Autoupgrade Non-CDB to PDB | Checking target database name in /etc/oratab file] ***
changed: [pilotdb] => (item=None)
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Autoupgrade Non-CDB to PDB | Fail if Target database name doesn't exist in oratab file.] ***
skipping: [pilotdb] => (item=None)
skipping: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Autoupgrade Non-CDB to PDB | Checking the status of the Target Container Database] ***
changed: [pilotdb] => (item=None)
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Autoupgrade Non-CDB to PDB | Fail if Target database is not up.] ***
skipping: [pilotdb] => (item=None)
skipping: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Autoupgrade Full DB | Generating response file for autoupgrade] ***
changed: [pilotdb] => (item=None)
changed: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Autoupgrade - Analyze] *******
```

```
changed: [pilotdb] => (item={'source_db_name': 'podba', 'source_db_home': '/u01/db12c/home', 'ora_home': '/u02/db19c/home', 'log_dir': '/u02/autoupgrade/dbupgrdlogs', 'start_time': 'NOW', 'restoration': 'no', 'upgrade_node': 'localhost', 'run_utlrp': 'yes', 'timezone_upg': 'yes', 'target_cdb_name': None, 'target_pdb_name': None})

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : debug] ***********************
ok: [pilotdb] => {
    "analyze.results[0].stdout_lines": [
        "AutoUpgrade 25.6.251016 launched with default internal options",
        "Processing config file ...",
        "+--------------------------------+",
        "| Starting AutoUpgrade execution |",
        "+--------------------------------+",
        "1 CDB(s) plus 2 PDB(s) will be analyzed",
        "Job 100 database podba",
        "+----+-------+---------+---------+-------+----------+-------+----------------+",
        "|Job#|DB_NAME|    STAGE|OPERATION| STATUS|START_TIME|UPDATED|         MESSAGE|",
        "+----+-------+---------+---------+-------+----------+-------+----------------+",
        "| 100|  podba|PRECHECKS|EXECUTING|RUNNING|  04:39:42| 1s ago|Executing Checks|",
        "+----+-------+---------+---------+-------+----------+-------+----------------+",
        "Total jobs 1",
        "",
        "Job 100 completed",
        "------------------- Final Summary --------------------",
        "Number of databases            [ 1 ]",
        "",
        "Jobs finished                  [1]",
        "Jobs failed                    [0]",
        "",
        "Please check the summary report at:",
        "/u02/autoupgrade/cfgtoollogs/upgrade/auto/status/status.html",
        "/u02/autoupgrade/cfgtoollogs/upgrade/auto/status/status.log"
    ]
}
PLAY RECAP *********************************************************************
pilotdb                    : ok=89   changed=50   unreachable=0    failed=0    skipped=35   rescued=0    ignored=0
```

SIHA Version after the upgrade:

```
-bash-5.1$ crsctl query has releaseversion
Oracle High Availability Services release version on the local node is [19.0.0.0.0]
-bash-5.1$ hostname
pilotdb
```

Reference output of Autoupgrade deploy mode:

```
[ansible@localhost playbooks]$ ansible-playbook upgrade_si.yml -i inventory.yml --ask-vault-pass --tags deploy Vault password:
[WARNING]: running playbook inside collection ibm.power_aix_oracle_dba

PLAY [pilotdb] *****************************************************************

TASK [Gathering Facts] *********************************************************
ok: [pilotdb]

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Autoupgrade - Deploy] ********
changed: [pilotdb] => (item={'source_db_name': 'podba', 'source_db_home': '/u01/db12c/home', 'ora_home': '/u02/db19c/home', 'log_dir': '/u02/autoupgrade/dbupgrdlogs', 'start_time': 'NOW', 'restoration': 'no',
'upgrade_node': 'localhost', 'run_utlrp': 'yes', 'timezone_upg': 'yes', 'target_cdb_name': None, 'target_pdb_name': None})

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : ansible.builtin.debug] *******
ok: [pilotdb] => {
    "deploy.results[0].stdout_lines": [
        "AutoUpgrade 25.6.251016 launched with default internal options",
        "Processing config file ...",
        "+--------------------------------+",
        "| Starting AutoUpgrade execution |",
        "+--------------------------------+",
        "1 CDB(s) plus 2 PDB(s) will be processed",
        "Job 101 database podba",
        "+----+-------+---------+---------+-------+----------+-------+----------------+",
        "|Job#|DB_NAME|    STAGE|OPERATION| STATUS|START_TIME|UPDATED|         MESSAGE|",
        "+----+-------+---------+---------+-------+----------+-------+----------------+",
        "| 101|  podba|PREFIXUPS|EXECUTING|RUNNING|  04:41:40| 0s ago|Executing fixups|",
        "+----+-------+---------+---------+-------+----------+-------+----------------+",
        "Total jobs 1",
        "",
.
.
.
.
        "+----+-------+---------+---------+-------+----------+-------+-------------------+",
        "|Job#|DB_NAME|    STAGE|OPERATION| STATUS|START_TIME|UPDATED|            MESSAGE|",
        "+----+-------+---------+---------+-------+----------+-------+-------------------+",
        "| 101|  podba|DBUPGRADE|EXECUTING|RUNNING|  04:41:40|14s ago|0%Upgraded CDB$ROOT|",
        "+----+-------+---------+---------+-------+----------+-------+-------------------+",
        "Total jobs 1",
.
.
.
.
        "+----+-------+---------+---------+-------+----------+-------+--------------------+",
        "|Job#|DB_NAME|    STAGE|OPERATION| STATUS|START_TIME|UPDATED|             MESSAGE|",
        "+----+-------+---------+---------+-------+----------+-------+--------------------+",
        "| 101|  podba|DBUPGRADE|EXECUTING|RUNNING|  04:41:40|36s ago|97%Upgraded CDB$ROOT|",
        "+----+-------+---------+---------+-------+----------+-------+--------------------+",
        "Total jobs 1",
        "",
        "+----+-------+---------+---------+-------+----------+-------+-------------------+",
        "|Job#|DB_NAME|    STAGE|OPERATION| STATUS|START_TIME|UPDATED|            MESSAGE|",
        "+----+-------+---------+---------+-------+----------+-------+-------------------+",
        "| 101|  podba|DBUPGRADE|EXECUTING|RUNNING|  04:41:40| 9s ago|0%Upgraded PDB$SEED|",
        "+----+-------+---------+---------+-------+----------+-------+-------------------+",
        "Total jobs 1",
.
.
.
.
        "+----+-------+---------+---------+-------+----------+-------+--------------------+",
        "|Job#|DB_NAME|    STAGE|OPERATION| STATUS|START_TIME|UPDATED|             MESSAGE|",
        "+----+-------+---------+---------+-------+----------+-------+--------------------+",
        "| 101|  podba|DBUPGRADE|EXECUTING|RUNNING|  04:41:40|30s ago|95%Upgraded PDBPODBA|",
        "+----+-------+---------+---------+-------+----------+-------+--------------------+",
        "Total jobs 1",
        "",
        "+----+-------+---------+---------+-------+----------+-------+--------------------+",
        "|Job#|DB_NAME|    STAGE|OPERATION| STATUS|START_TIME|UPDATED|             MESSAGE|",
        "+----+-------+---------+---------+-------+----------+-------+--------------------+",
        "| 101|  podba|DBUPGRADE|EXECUTING|RUNNING|  04:41:40| 0s ago|95%Upgraded PDB$SEED|",
        "+----+-------+---------+---------+-------+----------+-------+--------------------+",
        "Total jobs 1",
.
.
.
        "Total jobs 1",
        "",
        "Job 101 completed",
        "------------------- Final Summary --------------------",
        "Number of databases            [ 1 ]",
        "",
        "Jobs finished                  [1]",
        "Jobs failed                    [0]",
        "Jobs restored                  [0]",
        "Jobs pending                   [0]",
        "",
        "",
        "Please check the summary report at:",
        "/u02/autoupgrade/cfgtoollogs/upgrade/auto/status/status.html",
        "/u02/autoupgrade/cfgtoollogs/upgrade/auto/status/status.log"
    ]
}

TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Checking the status of the DBs] ***
changed: [pilotdb] => (item=None)
changed: [pilotdb]
```

```
TASK [ibm.power_aix_oracle_dba.podba_db_upgrade : Status of the DBs] ***********
ok: [pilotdb] => (item=podba) => {
    "msg": " Status: podba Database is running."
}

PLAY RECAP ***********************************************************************
    pilotdb                 : ok=5    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```